



Assignment - 4

Q. solve each of the following set of simultaneous congruences;

2. $x \equiv 1 \pmod{3}$, $x \equiv 2 \pmod{5}$, $x \equiv 3 \pmod{7}$

3. $x \equiv 5 \pmod{11}$, $x \equiv 14 \pmod{29}$, $x \equiv 15 \pmod{37}$

4. $x \equiv 5 \pmod{6}$, $x \equiv 9 \pmod{11}$, $x \equiv 3 \pmod{17}$

a. $x \equiv 1 \pmod{3}$, $x \equiv 2 \pmod{5}$, $x \equiv 3 \pmod{7}$

The moduli 3, 5, 7 are pairwise coprime
so CRT applies (Chinese Remainder Theorem)

1. compute total modulus: applies

$$m = 3 \times 5 \times 7 = 105$$

2. compute partial moduli:

$$m_1 = 105/3 = 35, m_2 = 105/5 = 21, m_3 = 105/7 = 15$$

3. Find inverses of m_i modulo each modulus:

- $35 \equiv 2 \pmod{3}$, inverse of 2 mod 3 is 2 (since $2 \times 2 = 4 \equiv 1 \pmod{3}$).
- $21 \equiv 1 \pmod{5}$, inverse is 1
- $15 \equiv 1 \pmod{7}$, inverse is 1.

4. combine

$$x \equiv (1 \times 85 \times 2 + 2 \times 21 \times 1 + 3 \times 15 \times 1) \\ = 70 + 42 + 45 = 157.$$

5. Reduce modulo 105:

$$157 \bmod 105 = 52$$

$$\therefore x \equiv 52 \pmod{105}$$

$$x \equiv 5 \pmod{11}, \quad x \equiv 19 \pmod{29}, \quad x \equiv 15 \pmod{31}$$

$$m = 11 \times 29 \times 31 = 9889$$

$$m_1 = 9889 / 11 = 899$$

$$m_2 = 9889 / 29 = 341$$

$$m_3 = 9889 / 31 = 319$$

Inverses:

$$899 = 8 \pmod{11} \rightarrow \text{inverse}(8 \bmod 11 = 7)$$

$$341 = 22 \pmod{29} - \text{inverse}(22 \bmod 29 = 4)$$

$$4 \times 22 = 88 \bmod 29 = 1$$

$$319 = 9 \pmod{31} \rightarrow \text{inverse } (9\%31) \text{ is } 7$$

$$\text{combine: } x = 5 \times 899 \times 7 + 14 \times 391 \times 4$$

$$+ 15 \times 519 \times 7$$

$$= 84056$$

$$\text{and, } 84056 \% 9889 = 9931$$

~~$$\text{given, } x \equiv 5 \pmod{6}, x \equiv 9 \pmod{11}, x \equiv 3 \pmod{17}$$~~

~~$$m = 6 \times 11 \times 17 = 1122$$~~

~~$$m_1 = 1122/6 = 187; m_2 = 1122/17 = 66$$~~

~~$$m_3 = 1122/11 = 102$$~~

$$\underline{\underline{c}}$$

$$x \equiv 5 \pmod{6}, x \equiv 9 \pmod{11}, x \equiv 3 \pmod{17}$$

$$1 \text{ } m = 6 \times 11 \times 17 = 1122$$

But note $6 = 2 \times 3$, not coprime with all others.

so first simplify using pairwise coprime system. handle modulo 6 as modulo 2 and 3.

from $x \equiv 5 \pmod{6}$:

$$x \equiv 1 \pmod{2}, x \equiv 2 \pmod{3}$$

Now combine $x \equiv 2 \pmod{3}, x \equiv 4 \pmod{11}, x \equiv 3 \pmod{17}$.

These moduli $(3, 11, 17)$ are coprime.

$$m = 3 \times 11 \times 17 = 561$$

$$m_1 = 561/3 = 187; m_2 = 561/11 = 51$$

$$m_3 = 561/17 = 33$$

Inverse.

$$\rightarrow 187 \bmod 3 = 1$$

$$\rightarrow 187 \bmod 11 = 7 \quad \text{inverse of } 7 \bmod 11 \text{ is } 8$$

$$\rightarrow 33 \bmod 17 = 16 \quad \text{inverse of } 16 \bmod 17 \text{ is } 16$$

$$\begin{aligned} \text{combine: } x &= 2 \times 187 \times 1 + 51 \times 4 \times 51 \times 8 + \\ &\quad 3 \times 33 \times 16 \\ &= 3590. \end{aligned}$$

$$x = (3590 \bmod 561)$$

$$= 224$$

224 is even, so add 561 to make it

$$\text{odd: } 224 + 561 = 785.$$